

Assignment 1: Fields & Waves (ECE230), Winter 2024

Total marks: 20. Deadline: 5 p.m. March 25, 2024. Submission: Dropbox outside B603, R&D Block (soft copies submitted to google classroom will be ignored)

Plagiarism policy: ZERO tolerance towards copying assignments from others/ plagiarism from any other sources. Such cases will be dealt strictly according to the institute policy.

Late submission policy: -5/day after the submission deadline (starts immediately after 5 p.m. March 25. NO exceptions).

***Vector quantities must be represented with an overhead arrow, vector calculus operators must be written correctly, dot and cross products must be written properly. In case of any of these mistake -1 will be awarded per question.*

NO credit will be given to the answers that do not accompany proper explanation.

Q1. Under what conditions can we have $\nabla^2 \vec{A} = -\mu_0 \vec{J}$? Prove the identity.

1 + 4 = 5 points

Q2. The vector potential of a single magnetic dipole $\vec{m}$ is given by:

$$\vec{A}(r) = \frac{\mu_0}{4\pi} \frac{\vec{m} \times \hat{r}''}{r''^2}$$

[Here, $\vec{r}'' = \vec{r} - \vec{r}'$]

Show that, for a distribution of magnetic dipoles, the total magnetic vector potential can be written in the following form (the proof is very similar to what we did the lecture for electric dipoles in obtaining surface and volume bound charges):

$$\vec{A}(r) = \frac{\mu_0}{4\pi} \left[\int_{V'} \frac{\vec{J}_b(r')}{r''} dv' + \oint_S \frac{\vec{K}_b(r')}{r''} dS' \right]$$

What does $\vec{J}_b$ and $\vec{K}_b$ represent?

10 points

Q3. An inhomogeneous medium is the one where the permittivity (ϵ) depends on the position (x, y, z) . Obtain $\vec{\nabla} \cdot \vec{E}$ for such a medium with a volume charge density $\rho(x, y, z)$.

5 points